

An Encoding Algorithm Based on the Shortest Path Problem

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Abstract - The transportation network of the city is dynamic and stochastic, The problem of dynamic stochastic shortest path is NP-hard. the optimal problem of path is widely used in the fields of transportation, communication and computer network. An improved self adaptive genetic algorithm is proposed by encoding the chromosomal mode. The paper investigates the shortest path problem based on the genetic algorithm principle, and improved genetic algorithm by adjusting the encoding parameters. Many experiments indicate that the improved genetic algorithm could adapt to new transportation rapidly in global optimization than A^* algorithm and Dijkstra algorithm and obtain the better solutions in the shortest path problem in the fields of transportation and computer network.

Keywords-Shortest path problem; encoding algorithm; chromosome encoding mode; optimization problems

I. INTRODUCTION

The Shortest path problem of path is widely used in the fields of computer network, communication and transportation, such as static deterministic optimal path problem and dynamic stochastic optimal path problem. The optimal path problem is one of classic combinatorial optimization problems in graph theory, it is stochastic and dynamic because of paths vary with time, and it is of great research and application value in the dynamic and stochastic optimal path problem [1-3]. The shortest path problem are on the basis of the path length, namely, there are many classic algorithms of shortest path problem, such as Floyd algorithm, Dijkstra algorithm, A^* algorithm and D^* algorithm, Two-tree Partitioning[4-7], and so on. The local search operators are inspired by Dijkstra's algorithm, the algorithm is carried out when the topology changes, in the population, it can generate local shortest path trees, which are used to promote the performance of the individuals. Genetic algorithm[8] and ant colony algorithm are several self-adaptive iterative algorithms[9]. Genetic algorithms are adaptive heuristic search algorithms, which are premised on the evolutionary ideas of natural selection and gene types. The basic concept of Genetic algorithms is designed to simulate processes in a natural system necessary for evolution, following the principles of survival of fittest in nature.

This research will use a genetic algorithm to optimize the parameters of the shortest path problem, optimizing mean finding an answer that is consistently accurate or satisfactory, though perhaps not the perfect answer to the problem. The goal is to really find some generally good parameters for the shortest path problem in the transportation network.

A. Optimization Problem

In static topologies, there are many deterministic algorithms for solving the shortest path problems, the shortest path problems are conventional combinatorial optimization problems. these deterministic algorithms are not efficient due to the necessity of restart in dynamic topologies. Nodes and paths are important elements in transportation network, the states of nodes and paths vary with time, although the states of transportation can be predicted according to the relevant path information, there are many problems between realistic states and predictive states. Some static deterministic algorithms of shortest path are not suitable for actual transportation network in big city [10-11].

The states of nodes and paths can follow probability distribution. Shortest path problem is to find the shortest path from source node to destination node in transportation network. The mathematical model of dynamic **stochastic shortest path problem may be described as follows:**

Suppose transportation network, $TN = \{V, E, W_V, W_E\}$, V is set of nodes and A is set of edges, W_V denotes the weight set of nodes, and W_E denotes the weight set of edges. The weights of nodes and edges represent the cost of vehicles on the node or path in the actual transportation network. **The cost W_V of node is a time-dependant discrete random variable**, while the cost W_E of edge is a stochastic process. The main distinction between dynamic stochastic shortest path problem and static deterministic shortest path problem is that not only the changes of over time of node cost are considered, but also the cost of statistic and probability distribution of edges are calculated [13-14].

The path is **minimum cost** and **does not contain any loop**, the problem of shortest path is NP-hard, The optimal solution of dynamic shortest path problem is the shortest one path chain from source node s to destination node t , it can't be solved in polynomial time. Genetic algorithm can effectively avoid broken roads and loop roads, and it is high-speed and flexible,

and it can meet the requirement of dynamic stochastic shortest path problem.

B. The Solutions of Encoding Genetic Algorithm

Several useful solutions for optimal combination problem and search problems can be generated by Genetic algorithm. Genetic algorithm is a **self adaptive search algorithm**, the algorithm simulates biological evolutionary and the mechanism of natural selection. Genetic algorithm is also the larger class of Evolutionary Algorithms, and it can generate solutions for optimization problems, such as **inheritance, mutation, selection, and crossover**, which are inspired by natural evolution. Along with a new form of crossover, one-point crossover, and point mutation, the concept of schema has been used to derive an improved schema theorem for genetic algorithm, which describes the schematic propagation from one generation to the next. The formal definition of genetic algorithm is described as follows:

$GA_{mc} = (E, F, P, S, O_c, O_m, T)$, E is the encoding mode, and it is commonly binary encoding mode, F is the fitness function, P is an initial population, S is the scale of the initial population, O_c is the crossover operator, O_m is the mutation operator and T is the termination condition.

II. THE CHROMOSOMAL ENCODING MODE

There are four shortcomings for the basic genetic algorithm in dynamic stochastic transportation network, the shortcomings are described as follows:

- According to peoples' experiences in basic genetic algorithm, the crossover operator and mutation operator are set so as to the two genetic operators can't be adjusted self adaptively.
- It is not suitable for the binary encoding mode in dynamic stochastic shortest path problem.
- The method generated easily leads to infeasible solutions for how initial population randomly, such as loop roads, duplicated nodes and broken roads.
- Such as no through traffic, no turning left and one way traffic, the operators of GA can't present the special states of paths in actual transportation network.

The encoding mode of chromosome in basic genetic algorithm is binary encoding mode. In transportation network, a feasible solution is consist of several nodes in a certain order. So real number encoding mode is more suitable than binary encoding mode.

It is binary encoding for the encoding mode of basic genetic algorithm. A solution is consist of several nodes in a certain order in transportation network, so real number encoding mode is more suitable than binary encoding mode. A gene represents a node in a chromosome, the **first gene is the source node s and the last one is the destination node t** . Given a transportation network which consists of n nodes, the chromosome length $L \in [2, n]$. we restrain each chromosome represents one path from source node s to destination node t without loops. So that each chromosome doesn't include

duplicated nodes and its length is at most n . The lengths of each chromosome do not have to be equal. we can encode according to chromosomes, the lengths of chromosomes are also different. Encoding like this not only represents the path length directly, but also decrease the scale of searching space effectively[15]. Suppose the topology graph of a transportation network is shown in Fig. 1.

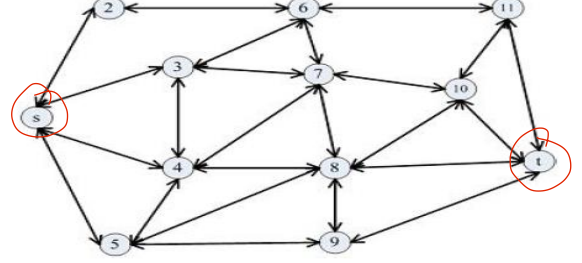


Fig.1 The topology graph of a transportation network

In Figure.1 there is a path from source node s to destination node t : $s \rightarrow 3 \rightarrow 7 \rightarrow 10 \rightarrow t$, the corresponding chromosome is shown in Fig. 2.

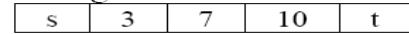


Fig. 2 The corresponding chromosome

A. The Problem of Initialization Population

First of all, set a weight on each arc according to the heuristic information such as **path length** and **residence time**. Then let **node s be the first gene of the chromosome**, we select one unlabeled node **adjacent to s** with the smallest weight and its out-degree **at least 2**, let this node be the **second gene**. In the same way, the path consist of selected nodes will extend to the destination node t , and we **check** whether this path contains any **loop** (duplicated nodes). If duplicated nodes exist, contract the duplicated nodes and all nodes between them into one node. Finally, we can gain one chromosome.

Suppose one path $s \rightarrow 4 \rightarrow 8 \rightarrow 9 \rightarrow 5 \rightarrow 8 \rightarrow t$ including duplicated nodes 8. Contract the sub-path $8 \rightarrow 9 \rightarrow 5 \rightarrow 8$ into one node 8 and finally we can gain the path $s \rightarrow 4 \rightarrow 8 \rightarrow t$.

The population initialization will be finished by repeat the above steps until reaching the specified number of chromosomes.

B. Strategy of Improved Selections

It is high probability for superior chromosomes to be selected and copied to the next generation in genetic algorithm. This improves the quality of entire population [16]. There are several selection strategies based on fitness function such as roulette wheel selection, tournament selection and $(\mu + \lambda)$ selection and so on. The chromosomes have higher probability to be selected to generate the next generation, which is larger function value. Unfortunately there are

shortcomings in the selection strategies mentioned above, there are faster all-pairs shortest paths via circuit complexity and combinatorial optimization in transportation network[17-18]. The fitness function of the k th chromosome is defined as follow:

$$f(k) = \sum E(W_V) + \sum E(W_E) \quad (2)$$

The improved selection strategy is a combination of roulette wheel selection and elitist selection.

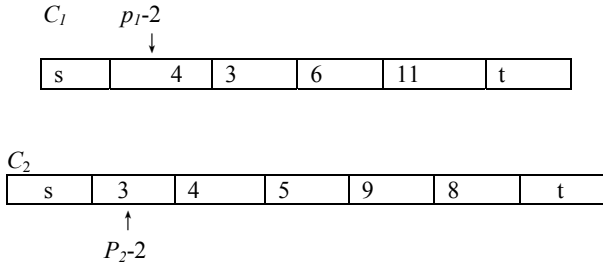
C. Strategy of Improved Crossovers

Suppose two chromosomes C_1 and C_2 , the improved crossover strategy is described as follows:

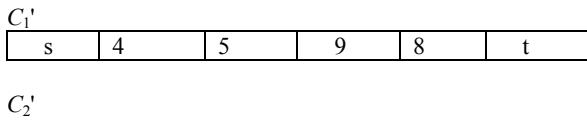
Improved Algorithm of Crossover

1. In chromosome C_1 and C_2 randomly, two crossover positions can be generated neither source node nor destination node.
2. Suppose the gene at position p_1 in chromosome C_1 is g_1 . Search for gene g_1 from position p_2 in chromosome C_2 .
3. All of genes between position p_1 and destination node t in chromosome C_1 will be replaced by those between gene g_1 and destination node t in chromosome C_2 .
4. If the same gene g_1 can't be found in chromosome C_2 , the genes in chromosome C_1 remain unchanged.
5. Suppose the gene at position p_2 in chromosome C_2 is g_2 , search for gene g_2 from position p_1 in chromosome C_1 .
6. All of genes between position p_2 and destination node t in chromosome C_2 will be replaced by those between gene g_2 and destination node t in chromosome C_1 .
7. If the same gene g_2 can't be found in chromosome C_1 , the genes in chromosome C_2 remain unchanged.

For example, chromosome C_1 and C_2 , crossover position $p_1=2$, $p_2=2$, shown in Fig. 3.



After crossover operation, the new chromosome C_1' and C_2' are shown in Fig. 4.



s	3	6	11	t
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Fig. 4 The Chromosome C_1' and C_2'

The new chromosomes after crossover operation maybe contain loops. We should ensure that the new chromosomes are minimum by contracting the duplicated nodes and all nodes between them into one node.

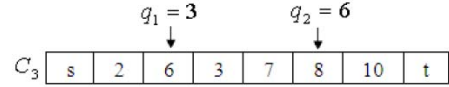
D. Strategy of Improved Mutations

The mutation operation of path is the procedure of one chromosome generated from another. The generated paths may be infeasible solutions without any constraint. Therefore it is necessary to improve the mutation strategy. The improved mutation strategy is described as follows:

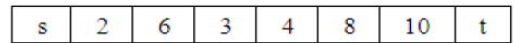
Improved Mutation Algorithm

- Step1. Two mutation positions q_1 and q_2 can be generated in chromosome randomly.
- Step2. Suppose the value at position q_1 of gene is x_1 , and the value position q_2 of gene is x_2 .
- Step3. A path between x_1 and x_2 can be generated randomly.

For example, chromosome C_3 , mutation position $q_1=3$ and $q_2=6$, shown in Fig. 5.



The corresponding nodes at mutation position q_1 and q_2 are node 6 and node 8. Generate a new subpath 6→3→4→8 from node 6 to node 8 randomly. The chromosome after mutation operation is shown in Fig. 6.



E. Genetic Operators of Self-adaptive Adjustment

There are two important operators in genetic algorithm, one is crossover operator $O_c \in [0.5, 0.9]$ and another is mutation operator $O_m \in [0.1, 0.3]$. The crossover operation implements the process of combinatorial optimization while the mutation operation implements the process of random search. If the next generation of population is not better than previous generation of population, mutation operator O_m

should be increased and crossover operator O_c should be decreased. Conversely, mutation operator O_m should be decreased and crossover operator O_c should be increased. Suppose the genetic algorithm has evolved to the n th generation of population, the relationship between crossover operator O_c and n is:

$$O_c = \text{crossover}(n) = \text{crossover}(n-1) + \text{sgn}(\Delta f \times C_1 \times e^{-\Delta f}) \quad (3)$$

Where c_1 is a constant factor, $\text{crossover}(0)=0.75$, Δf is the difference of fitness function value between the n th generation and the $(n-1)$ th generation. The relationship between mutation operator O_m and n is:

$$O_m = \text{mutation}(n) = \text{mutation}(n-1) - \text{sgn}(\Delta f \times C_2 \times e^{-\Delta f}) \quad (4)$$

Where c_2 is a constant factor, $\text{mutation}(0)=0.23$, Δf is the difference of fitness function values between the n th generation and the $(n-1)$ th generation.

III. THE ALGORITHM OF IMPROVED ENCODING GENETIC

The steps of the improved encoding genetic algorithm are depicted as follows:

Gene Algorithm

1. Initialize the weight of W_v of each node and the weight of W_E of each arc
2. Initialize the population.
3. Calculate the fitness function value $f(k)$ of each chromosome in the population.
4. Adjust crossover operator O_c and mutation operator O_m .
5. Crossover operation.
6. Selection operation
7. Mutation operation.
8. The algorithm stops, otherwise, goto 3. if reaching the termination condition,

The problem in this algorithm is that according to the initial population, the improved genetic algorithm would make the program to run under a large search space with small space complexity. Sometimes the individuals may rapidly come to dominate the population, causing it to converge on a local maximum as it is common for many GAs. Once the population has converged, the ability of the GA to continue to search for better solutions is effectively eliminated.

Results obtained from the research is given and discuss how the variations of the parameters which affect the end result. Content of the research with further extensions is concluded. These research is to propose a new method to solve the shortest path problem using genetic algorithms. The solution aims to achieve an increased number of successful and valid convergence using evolutionary computing techniques. Our experimental results show that this algorithm finds more than one possible solution for a given source and destination and this makes it easy to find the next shortest path which exists other than the optimal solution.

IV. CONCLUSION AND EXPERIMENTS

The Encoding algorithm has been tested for a map containing more than 20000 nodes. In most situations the initial population may contain routes that are impossible to traverse in practical situations. The selection criteria for the new generations are totally based on the fitness value given to each individual. But the changes in probabilities of recombination operators may produce surprising results. It is convenient to keep the operators' values in a certain acceptable range.

The aim of simulation experiments is to find the shortest path from node s to node t using *DRSP-GA*, Dijkstra and *A** algorithm separately. The experimental results guarantee to provide good solutions acceptable for the given search space. The results are shown in TABLE I.

TABLE I
THE COMPARATIVE OF THREE ALGORITHMS

The Number of Nodes	The Number of Arcs	DRSP-GA		Dijkstra		A*	
		Avg of Time	Avg of Shortest Path	Avg of Time	Avg of Shortest Path	Avg of Time	Avg of Shortest Path
1000	10000	0.005	1037	0.013	1037	0.011	1037
1500	15000	0.007	1409	0.037	1409	0.019	1409
2000	20000	0.011	1901	0.046	1885	0.032	1885

The simulating results of the experiment show that the improved genetic algorithm proposed by this paper is effective, although failed to get the optimal solution occasionally, and the algorithm has much higher capacity of global optimization than algorithm Dijkstra and algorithm *A** in the dynamic stochastic shortest path problem in the scale of transportation network.

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The dynamic stochastic shortest path problem and its mathematical model are introduced. The simulation experimental results show that the improved genetic algorithm proposed by this paper has much higher capacity of global optimization than Dijkstra and *A** algorithm in the shortest path problem. The research is supported by Chinese Natural Science Foundation (61672328), the Science and Technology Development Planning Project of Shandong Province (2014GGX101029), the Science and Technology Project Plan of Ministry of Housing and Urban-Rural Development of P.R.C (2014-K8-071), the Doctor Foundation of Shandong Jianzhu University (XNBS1438).

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